👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3A

Determine Equivalent Ratios Using Graphs

Lesson 3-4 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the values from a ratio table as points on the coordinate plane.

Language Objective: I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Coordinate plane | **Ordered pair** | **Linear pattern** | **Proportional** | **Origin** | **Ratio table** | **Unit rate point** | **Interpret** | **Coordinates**

Level 3: try the blanks from memory first, then check the bank.

1. The flat grid formed by a horizontal and a vertical number line is the _____.

2. A set of two numbers (x, y) that names a point is an _____.

3. When graphed points form a straight line, they show a _____ .

4. Two quantities that always keep the same ratio are _____ .

5. The point (0, 0) where the two axes meet is the _____ .

6. A table that lists equivalent ratios in rows or columns is a _____ .

7. The point (1, r) on a ratio graph is the _____ .

8. To say what a point means in words is to _____ it.

9. The x-value and y-value of a point are its _____ .

Write About the Math

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Coordinate plane** — *Plano cartesiano*
- ☐ **Ordered pair** — *Par ordenado*
- ☐ **Linear pattern** — *Patrón lineal*
- ☐ **Proportional** — *Proporcional*
- ☐ **Origin** — *Origen*

Model: *The points form a straight line, and each sundae adds 2 ounces of sauce.* — Listen for 'straight line through the origin' and the connection that the constant 2-ounce step shows a proportional relationship.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Every point on the line has the same ratio — ___ taco to ___ ounces of salsa — so 10 tacos would use ___ ounces.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Coordinate plane
2. **Notes 2:** Ordered pair
3. **Notes 3:** Linear pattern
4. **Notes 4:** Proportional
5. **Notes 5:** Origin
6. **Notes 6:** Ratio table
7. **Notes 7:** Unit rate point

8. **Notes 8:** Interpret
9. **Notes 9:** Coordinates
10. **Watch for this mistake:** *A common mistake in Graph Ratio Tables is writing the ordered pair in the wrong order — putting the second table value first instead of matching (x, y) to the table's column order. For example, if a recipe table shows 2 cups of sugar for every 4 cups of flour, a student might plot (4, 2) instead of (2, 4), landing the point in a completely different spot on the grid. Before you plot, check: which quantity goes on the x-axis, and does your ordered pair start with that number?*
11. **Try It 1:** A. (3, 6) — *Cups (x) come first, scoops (y) second. 3 cups need 3 × 2 = 6 scoops, so the point is (3, 6).*
12. **Try It 2:** A. y = 2x — *Each cup needs 2 scoops, so y (scoops) is always 2 times x (cups): y = 2x. For 4 cups, y = 2 × 4 = 8.*